

Effects of Font Type and Weight on Reading in VR

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ABSTRACT

This study focused on font type and weight and we examined their effects on the readability of long texts in VR. In the experiment, we changed the font weight of three font types, Gothic, Mincho and Antigothic, to evaluate the readability of text with a length of 600 characters. Our results showed a tendency for Antigothic to be more readable and less fatiguing than Mincho and Gothic.

CCS CONCEPTS

• Human-centered computing → Virtual reality; Empirical studies in HCI; User studies.

KEYWORDS

Virtual reality, Text readability, E-Book

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1 INTRODUCTION

With the spread of virtual reality (VR), educational materials using VR have been developed[Corporation 2019]. VR educational materials are a good aid in learning interactively and most of these materials provide immersive 2D/3D CG animations and videos. However, there is a disadvantage that students cannot see anything that exists in the real space while wearing a head mounted display (HMD) to experience VR. So, when they want to read their text books in the real space, they have to take off the HMD every time. Displaying long texts, such as sentences in text books, in the VR space is desired in the educational use of VR to bring further learning effects of presenting sentences and these materials in the VR space at the same time.

There have been many studies on the readability of long texts using common mobile displays, such as tablets[Rello et al. 2016]. On the other hand, as regarding VR spaces, most of the research on the readability of texts on HMDs is of words, that is, not long texts[Büttner et al. 2020], and there is still little research on the readability of sentences. Therefore, we have studied the readability of long texts in a VR space, using sentences in high school and junior high school text books[Kobayashi et al. 2021b][Kobayashi

et al. 2021a]. The results suggested that text with a font size of 18 pt to 22 pt (1.20° to 1.40° visual angle) is easier to read at a viewing distance of 30 cm.

In a previous study [Kobayashi et al. 2021b], the effect of font on readability was also examined. We found that the font had an effect on fatigue. However, detailed interpretation of the results was difficult due to various factors such as font type. Therefore, it is necessary to conduct another survey regarding the font type. In addition, we had comments from some of the subjects that the thickness of letters influenced on fatigue, so it is necessary to consider the thickness of letters as well. In this study, we examine the readability of long texts in a VR space by focusing on the font type and weight (thickness of letters).

2 EXPERIMENT

2.1 Environment and participants

The participant sat in a chair and wore an HMD (HTC, Vive Pro2) while holding a controller (HTC, included in Vive Pro2) in his hand. The participant was asked to lean against the back of the chair to maintain a constant viewing distance from the text. The participants were 22 male and 4 female university students (age range: 21-24 years).

2.2 Stimuli

The font types used for the text are Mincho, Gothic, and Antigothic (Antique for kana and Gothic for Kanji). Mincho and Gothic are font types used in books and newspapers, while Antigothic is mainly used in manga in Japan. All of the font types were chosen as stimuli because they are considered to be the font types that the participants are likely to see frequently. We used the fonts: Yu Mincho, Yu Gothic (Jiyu Kobo Ltd.), and Antique Cezanne (Fontworks Co., Ltd.), respectively. Two types of font weight, regular and bold, were provided for each of three font types, so we had the six stimulus conditions in total. Figure 1 shows examples of the six stimulus conditions.

The viewing distance from the text was fixed at 30 cm. The image of the text was created in A5-size, and the participant was presented with two A5-size texts side by side in an A4-size spread. The top and bottom margins of each page were 14 mm, and the left and right margins were 12 mm. The font size was 22 pt, the line spacing was 44 pt, the background color was white [RGB(255,255,255)], and the font color was black [RGB(0,0,0)]. These conditions were set based on the results of the previous studies[Kobayashi et al. 2021b][Kobayashi et al. 2021a].

The sentences across 4 pages were approximately 600 characters. In order to achieve some uniformity in the level of sentences, the sentences were extracted from a Japanese language problem book for junior high school students.

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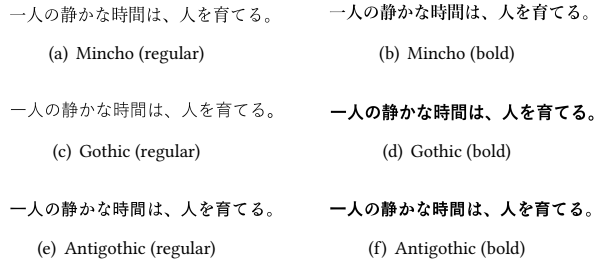


Figure 1: Examples of the six stimulus conditions

2.3 Procedure

A sentence from one of the six stimulus conditions was presented in the VR space. The participant started reading silently by pressing the trigger button on the controller. The participant could switch pages by pressing the controller's trackpad and read through the full sentences. When all the reading was completed, the trigger button on the controller was pressed to end the silent reading. The time between the first press and the second press of the trigger button was defined as the reading time of the sentences.

After reading silently, the participant pressed the grip button on the controller to display the fill-in-the-blanks questions and started answering them orally at the same time. The fill-in-the-blank questions were provided to ensure that the participant had read and understood the sentences. The fill-in-the-blank questions were presented in the same font type and weight as the sentences presented during silent reading. There were three questions in total. When the answering was completed, the participant pressed the grip button to finish answering the questions. The time between the first press and the second press of the grip button was defined as the answering time of the questions.

After answering the questions, the participant took off the HMD and filled out the evaluation sheet. On the evaluation sheet, the participant was asked to answer whether the participant have read the presented sentences before and to rate the readability and fatigue on a seven-point scale from 1 to 7.

One trial was from silent reading to filling in the evaluation sheet, and a total of six trials were performed in random order by changing the font type and weight. The sentences in the six trials were different.

3 RESULTS AND DISCUSSION

Twelve data points from the two participants who were incomplete during the experiment, those who had previously read the experimental stimulus sentences, and those who answered less than two-thirds of the fill-in-the-blank questions correctly were excluded from the analysis. The final number of data used in the analysis was 144 points.

Since the normality was not found in the data, Friedman's tests were conducted with the readability and fatigue as dependent variables and Mincho (regular), Gothic (regular), and Antigothic (regular) as independent variables. The results showed no significant difference between the font types. However, when looking carefully at the box-and-whisker diagrams of the evaluation as shown in

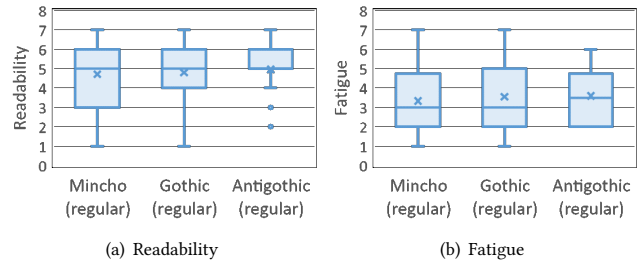


Figure 2: Results under the regular weight condition

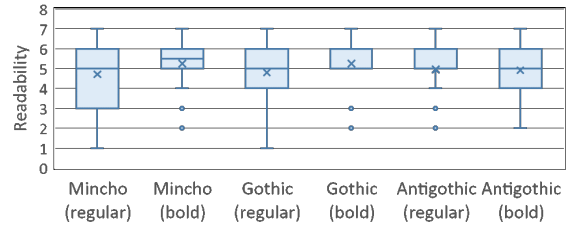


Figure 3: Results of the readability

Figure 2, the range of the evaluation rates has less variation for the font type Antigothic. It shows a tendency for Antigothic to be more readable and less fatiguing than Mincho and Gothic. These days, e-books including manga have become familiar with the young generations, so that the number of opportunities to see Antigothic font type in manga on the display has increased and it is possible for young people to be more accustomed to reading long texts displayed on the HMD in Antigothic font type than in Mincho and Gothic font types.

To compare the evaluation of the readability and fatigue by the font weight, Wilcoxon signed rank tests were conducted between regular and bold weight conditions for Mincho, Gothic and Antigothic font types, respectively, and no significant difference was found. Figure 3 shows the evaluation of the readability under each weight condition. As shown in Figure 3, the bold condition had consistently higher readability for Gothic and Mincho. On the other hand, Antigothic was evaluated higher under the regular condition. In general, it is easier to read long texts when the letters are not so strongly impressed[Tsutawaru 2018]. Antigothic font type under the bold condition could have been too bold for reading long texts.

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